



CN Lab: Assignment

Instructor: Muhammad Nouman Hanif

Date: September 26, 2025

This assignment is for practice and will not be graded.

Instructions:

- Write your name and roll number in the space provided.
- Answer all questions.

Name: Ahsan Baig. **Roll No.:** 23L-0515.

Assignment Questions

Question # 01

- (a) Open the Wireshark(1).pcap file on Wireshark. Apply HTTP filter and select the packet having serial no. 1338 and answer the following questions:
- 1) What is the expiry and last modified date and time?
A) Expires: Fri, 04 Aug 1978 12:00:00 GMT. Last-Modified: Mon, 18 Jun 2007 22:47:40 GMT.
 - 2) What is the TCP segment length and payload length?
A) TCP segment length: 329. Payload length: 35.
 - 3) What is the source and destination port number?
A) Source port: 80. Destination port: 2546.
- (b) Answer the following questions:
- 1) Why is a switch preferred over a hub in modern networks?
A) Switch is preferred because it forwards frames only to the intended device (unicast), reducing collisions, improving bandwidth efficiency, and enhancing security, unlike a hub which broadcasts to all ports.
 - 2) Your assigned IP address follows this format: 76.x.29.y where, x is the first two digits of your roll number and y is the last two digits of your number. Based on your own IP address, determine:
 - Network Address → 76.x.29.0
 - Broadcast Address → 76.x.29.255
 - 4th Host Address → 76.x.29.4

Question # 02

(a) Open the Wireshark(2).pcap file on Wireshark. Apply DNS filter and select the packet having serial no. 8 and answer the following questions:

1) What is the destination UDP port number and the source MAC address?

A) Destination UDP port number: 4375. Source MAC address: **00:16:b6:f4:eb:a8**.

2) How many “answers” are provided in the DNS message? What is/are the answer(s)?

A) 1 answer; Answer: www.mit.edu: type A, class IN, addr 18.7.22.83.

3) What domain name is being queried in the DNS packet?

A) www.mit.edu.

(b) Answer the following questions:

1) How will you differentiate between client and server errors? Give example?

A) Client errors are **4xx** status codes. Server errors are **5xx** status codes.

2) IP addresses are given as follows:

- 126.25.80.60
- 195.60.75.150
- 191.85.15.0
- 225.10.5.200

For each IP address, calculate the following:

- Network address.
 - Broadcast address.
 - Host address.
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- **126.25.80.60 (Class A /8):** Network = 126.0.0.0, Broadcast = 126.255.255.255, Host = 126.25.80.60.
 - **195.60.75.150 (Class C /24):** Network = 195.60.75.0, Broadcast = 195.60.75.255, Host = 195.60.75.150.
 - **191.85.15.0 (Class B /16):** Network = 191.85.0.0, Broadcast = 191.85.255.255, Host = 191.85.15.0.
 - **225.10.5.200 (Class D):** Multicast address – no network/broadcast/host in the unicast sense.

Question # 03

(a) Open the Wireshark(3).pcap file on Wireshark. Apply DNS filter and select the packet having serial no. 1127 and answer the following questions:

1) What is time when first frame was received?

A) Arrival Time: Jun 29, 2007 07:05:40.039050000 PKT.

2) What is the destination port number and source MAC address?

A) Destination port: 53. Source Mac address: **00:13:02:d1:b6:4f**.

3) What is the domain name being queried in the selected packet?

A) www.google-analytics.com

(b) Answer the following questions:

1) Describe the range of IP addresses for private address space.

2) IP addresses are given as follows:

- 100.200.50.25
- 220.110.90.1
- 198.51.100.75
- 72.1.1.1

For each IP address, calculate the following:

- Network address
- Total Number of Hosts.
- Reverse the IP and calculate both of the above.

A) **100.200.50.25** – Network 100.0.0.0 (/8), usable hosts 16,777,214. Reversed 25.50.200.100 – Network 25.0.0.0 (/8), usable 16,777,214. 220.110.90.1 – Network 220.110.90.0 (/24), usable hosts 254. Reversed 1.90.110.220 – Network 1.0.0.0 (/8), usable 16,777,214. 198.51.100.75 – Network 198.51.100.0 (/24), usable hosts 254. Reversed 75.100.51.198 – Network 75.0.0.0 (/8), usable 16,777,214. 72.1.1.1 – Network 72.0.0.0 (/8), usable hosts 16,777,214. Reversed 1.1.1.72 – Network 1.0.0.0 (/8), usable 16,777,214.

Question # 04

(a) Open the Wireshark(4).pcap file on Wireshark. Apply TCP filter and select the packet having serial no. 573 and answer the following questions:

1) What is sequence number and arrival time of the selected packet?

A) Sequence number: 26594. Arrival time: Jun 29, 2007 07:05:32.019018000 PKT.

2) What is the TCP segment length and payload length?

A) Segment length: 1460 bytes. Payload: 1460 bytes.

3) What is the source and destination MAC address?

A) Source: 00:16:b6:f4:eb:a8. Destination: 00:13:02:d1:b6:4f.

(b) Answer the following questions:

1) Why is a switch preferred over a hub in modern networks?

A) Switch is preferred because it forwards frames only to the intended device (unicast), reducing collisions, improving bandwidth efficiency, and enhancing security, unlike a hub which broadcasts to all ports.

2) IP addresses are given as follows:

- 10.50.50.1
- 192.168.5.1
- 221.72.20.0
- 167.0.113.200

For each IP address, provide the following:

- Network address.
- Broadcast address.
- Reverse the IP and determine:
 - Class of the IP?
 - If the IP is a part of private address space or not?

A) **0.50.50.1 (Class A/8)** Network: 10.0.0.0; Broadcast: 10.255.255.255. Reversed 1.50.50.10 → Class A (1.x), **Not private.** **192.168.5.1 (Class C/24)** Network: 192.168.5.0; Broadcast: 192.168.5.255.

Reversed 1.5.168.192 → Class A (1.x), **Not private.** **221.72.20.0 (Class C/24)** Network: 221.72.20.0; Broadcast: 221.72.20.255. Reversed 0.20.72.221 → First octet 0 (reserved/special), **Not private.**

167.0.113.200 (Class B/16) Network: 167.0.0.0; Broadcast: 167.0.255.255. Reversed 200.113.0.167 → Class C (200.x), **Not private.**

Best of Luck